

LAB 3: Cellular Communication System

Aim

- To develop and analyze a MATLAB-based cellular communication system simulator

Objective

- Visualize cellular coverage regions using hexagonal cells.
- Analyze received signal strength (RSSI).
- Simulate user mobility in a cellular environment.

Components & Equipment

- MATLAB – Online or offline version.

Theory

A cellular communication system divides a geographical region into smaller areas known as cells. Each cell is served by a base transceiver station (BTS), allowing efficient reuse of frequencies and increased network capacity.

Modern wireless communication systems such as GSM, LTE, and 5G use cellular architecture to support voice communication, data transmission, mobility management, and seamless handover between cells.

This laboratory implements a MATLAB-based simulator that visualizes:

- cellular layout,
- user movement,
- signal coverage,
- communication routing,
- and interference effects.

Cellular Concept

A cellular system divides a service area into hexagonal cell where each cell contains Base Transceiver Station (BTS) and Users/Mobile Stations (MS).

The advantages of using cellular systems include:

- Frequency reuse
- Increased capacity
- Reduced transmission power
- Efficient mobility management

Hexagonal Cell Structure

Hexagonal cells are used because they provide continuous coverage, minimize overlap, and provide approximate circular radiation patterns.

Frequency reuse allows the same frequencies to be reused in non-adjacent cells.

Cluster size:

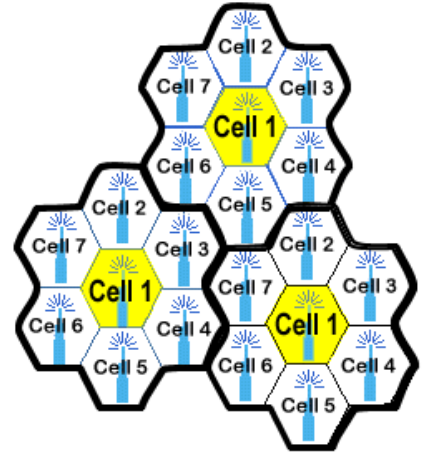
$$N = i^2 + ij + j^2$$

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Where:

- N = cluster size
- i, j = shift parameters

The common cluster sizes are 3, 4, 7, and 12



Signal Propagation and Path Loss

Signal strength in cellular communications decreases with distance due to path loss.

Received signal strength:

$$P_r = P_t - 10n \log_{10}(d)$$

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Where: P_r = received power (dBm), P_t = transmitted power (dBm), n = path loss exponent, and d = distance between user and BTS

Typical values of n are Free space: 2, Urban: 3–4, and Indoor: 4–6

A mobile user connects to the BTS with the strongest received signal. This is based on Association criterion:

$$\text{Best BTS} = \text{Maximum RSSI}$$

Handover occurs when a mobile user moves from one cell to another base on the following condition:

$$\text{If } \text{RSSI}_{\text{neighbor}} > \text{RSSI}_{\text{current}} \rightarrow \text{Handover occurs}$$

The main purpose of handover is to maintain communication continuity and avoid dropped connections.

SINR (Signal to Interference plus Noise Ratio) determines communication quality.

$$\text{SINR} = \frac{S}{I + N}$$

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Where: S = desired signal power, I = interference power, and N = noise power

Higher SINR means better communication quality and higher throughput.

Core Network Routing

The core network performs switching, routing, and mobility management.

Its communication path follows this pattern:

$$\text{User A} \rightarrow \text{Serving BTS} \rightarrow \text{Core Network} \rightarrow \text{Destination BTS} \rightarrow \text{User B}$$

Requirements

- Personal Computer
- Software: MATLAB R2020 or later (online or offline versions)
- Optional Toolboxes:
 - Communications Toolbox
 - DSP Toolbox
- Files Required
 - main.m

- createCells.m
- drawHexagon.m
- drawNetwork.m
- createUsers.m
- moveUsers.m
- computeSignal.m
- assignBaseStation.m
- handoverCheck.m
- animatePacket.m
- drawRSSIHeatmap.m
- drawSINRMap.m
- drawTrafficStats.m

Procedure

- i. Open MATLAB (offline version) or log in (online version)
- ii. Create the project folder.
- iii. Save all MATLAB files in the folder.
- iv. Run the main.m
- v. Observe:
 - a. hexagonal cells,
 - b. BTS towers,
 - c. cluster arrangement,
 - d. frequency reuse colors.
- vi. Observe user movement and identify serving BTS, user trajectories, and mobility patterns.
- vii. Record:
 - a. number of users,
 - b. connected cells.
- viii. Open RSSI heatmap window and observe strong signal regions, weak signal regions, and cell-edge areas.
 - a. Change path loss exponent n and transmit power P_t and record changes in coverage area and RSSI distribution
- ix. Open SINR map window and observe signal quality variations across the network, including high-quality central regions and low-quality edge regions.
 - a. Identify interference-prone areas and regions affected by co-channel interference from neighboring cells.
 - b. Change the path loss exponent n and transmit power P_t , and observe changes in SINR distribution, interference intensity, and overall network quality.
- x. Observe mobile users in the cellular topology window moving across different cells and switching between base stations. Monitor handover events displayed in the command window and identify when users change serving BTS due to stronger signal availability. Change user speed, movement patterns, and cell density, and observe changes in the number of handovers and mobility behavior.

- xi. Using the cellular topology window and observe end-to-end communication between source and destination users through the network. Follow packet movement from source user → serving BTS → core network → destination BTS → destination user and change source and destination user positions and observe variations in routing paths and packet travel distance.
- xii. Open traffic statistics window and observe user distribution across different cells and the corresponding load per base station.
 - a. Identify heavily loaded and lightly loaded cells and analyze network congestion behavior.
 - b. Change number of users and user mobility patterns, and observe changes in traffic distribution, cell congestion, and load balancing across the network.
- xiii. Record the results in table 1 below

Table 1

Parameter	Observation
Number of cells	
Number of users	
Frequency groups	
Maximum RSSI	
Minimum RSSI	
Highest SINR region	
Lowest SINR region	
Number of handovers	
Most congested cell	

MATLAB Code

Check the attached files

Questions

1. Repeat the experiment using a 12-cell arrangement
2. Define co-channel interference and explain how frequency reuse leads to interference between cells using the same frequency band.
3. Explain how movement of users influences signal strength variation, handover frequency, routing complexity, and network resource allocation.
4. Describe what the path loss exponent represents and how it affects signal attenuation with distance

Submission

- The work is to be done in groups

- The lab report must contain completed tables, necessary calculations, and Network design explanation